

Progressive Retrieval Attention on OpenVINO GenAI: Semantic Resource Identity, Continuous-Batching Evidence, and the Native K/V Boundary

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Abstract

OpenVINO GenAI provides continuous batching, bounded K/V scheduling, prefix caching, eviction controls, and execution across Intel CPU, GPU, and NPU devices. Progressive Retrieval Attention (PRA) adds semantic, query-addressed resources that must remain distinct from sequential prefixes. We implement an OpenVINO provider, an E0 selected-text adapter, and tested contracts for topology-bound native K/V handles, authorization, request pinning, exactly-once attachment, cleanup, and prefix-pool exclusion. A live audit of OpenVINO 2026.3.1 and OpenVINO GenAI 2026.3.1.0 confirms all required continuous-batching controls, but finds no public operation for attaching arbitrary selected non-prefix K/V. We therefore report native E2 as blocked, not as an implicit consequence of prefix caching. On Qwen2-0.5B-Instruct INT4, selected-text and full-context E0 recover the controlled answer in every evidence-bearing request on an Intel i7-1355U CPU and Iris Xe GPU. Selected text uses 63 input tokens instead of 1,573. On GPU it reduces cold TTFT from 1,129.6 to 102.9 ms. Under explicitly warmed continuous batching at eight shared requests, it reaches 8.48 requests/s rather than 7.39 and reduces TTFT p99 from 297.7 to 148.9 ms. The result establishes a reproducible transport baseline and a precise extension boundary; it does not establish native OpenVINO PRA parity or memory savings.

1 Introduction

An inference engine’s prefix cache identifies previously computed sequential token histories. PRA identifies immutable logical resources selected by a query. These identities have different lifetimes and authorization rules:

$$I_{\text{prefix}} \neq I_{\text{PRA}} \neq I_{\text{physical}}.$$

Identical visible prompts may select different protected resources, while the same resource may be shared safely by many sessions. A correct OpenVINO integration must preserve that distinction while allowing OpenVINO to keep control of compiled execution, K/V blocks, eviction, and batching.

This study makes three contributions. It implements the PRA-side OpenVINO contracts, audits the installed GenAI scheduler and pipeline interfaces, and measures a common E0 workload across CPU, GPU, prefix reuse, and continuous batching. The audit also identifies the smallest missing native mechanism: request attention metadata cannot currently receive an arbitrary selected non-prefix K/V set through the public Python API.

2 Claim Boundary

E0 renders selected source text into the visible prompt. E1 may retain derived engine state without exposing arbitrary selected K/V to attention. E2 consumes that K/V natively, preserving source geometry and combining local and memory scores in one softmax. All model measurements in this paper are E0. The native classes described below are executable lifecycle contracts, not evidence that OpenVINO already performs E2.

3 OpenVINO Mapping

3.1 Responsibility split

PRA owns typed records, selection, authorization, task/session policy, semantic retention, and a four-tier storage lifecycle. OpenVINO owns compilation, device execution, sequential K/V blocks, prefix reuse, batching, and physical eviction. Physical tensor handles are derived state bound to a specific model, revision, device, precision, layer count, K/V-head count, and head dimension.

3.2 Implemented code

`src/pras/openvino/adapter.py` provides `OpenVINOEngineAdapter`. Its HTTP/OVMS form is explicitly E0. It reports E2 only when an application supplies an `OpenVINONativeExecutor`; package installation or prefix-cache availability is never enough.

`src/pras/openvino/native.py` provides a topology fingerprint, immutable K/V handles, a physical-store protocol, an in-memory deterministic store, and `OpenVINONativeAttachmentManager`. The manager validates topology before registration, authorizes a complete routed set before scheduler admission, pins it for the request lifetime, returns it exactly once, and unpins it during cleanup. A shared isolation guard rejects any PRA logical key offered to the ordinary sequential/prefix pool. Focused tests cover topology mismatch, cross-tenant access, duplicate attachment, cleanup, and capability honesty.

3.3 Live API boundary

`SchedulerConfig` exposes cache size or K/V-block count, sequence and batched-token limits, prefix caching, dynamic split/fuse, and cache eviction. Supplying this configuration activates continuous batching. The public `LLMPipeline` surface exposes generation, tokenizer/config access, and chat state. It does not expose selected external K/V, non-prefix tensor attachment, or an attention-metadata callback. Prefix caching therefore accelerates repeated visible histories but does not implement PRA E2.

4 Experimental Method

4.1 Environment

The remote Windows host runs Python 3.10.0, OpenVINO 2026.3.1, OpenVINO GenAI 2026.3.1.0, and OpenVINO Tokenizers 2026.3.1.0. Available devices are a 13th Gen Intel Core i7-1355U CPU and Intel Iris Xe integrated GPU; no NPU is available. The model is OpenVINO Qwen2-0.5B-Instruct INT4. Scheduler settings use a 1 GiB cache, eight sequences, 2,048 batched tokens, prefix caching, and dynamic split/fuse. Greedy decoding is capped at 24 new tokens.

4.2 Conditions

The target resource contains `PRA_EVIDENCE_4821`; distractors contain different codes. The five one-shot conditions are no evidence, stable prefix only, selected text, prefix plus selected text, and full retained context. The same evidence is used in selected and full conditions. Each condition is run five times independently on CPU and GPU.

The direct `ContinuousBatchingPipeline` experiment uses selected text and full context under shared-resource and independent-resource workloads. Concurrency is 1, 2, 4, and 8 with five measured waves. Every distinct resource is primed outside the timed interval. Reported p95/p99 values are smoke descriptors over small cohorts, not production service-level estimates.

5 Results

5.1 Capability status

Table 1: OpenVINO integration status after live package and hardware audit.

Capability	Status	Evidence
Selected-text E0	Ready	CPU/GPU live generation
Prefix cache controls	Ready	live SchedulerConfig audit
Continuous batching	Ready	direct batched generation
PRA lifecycle/isolation	Contract ready	focused tests
Native selected K/V E2	Blocked	no public attachment operation
OVMS E0	Adapter ready	endpoint not installed on host
NPU comparison	Not run	no NPU device available

5.2 CPU and GPU one-shot behavior

Table 2: Evidence-bearing E0 conditions. Visible is mean input tokens; times are milliseconds. Success is controlled answer recovery.

Device	condition	visible	success	cold TTFT	warm TTFT	latency p50
CPU	pra_only	63	1.00	549.2	336.8	1641.2
CPU	prefix_plus_pra	361	1.00	909.8	191.2	1118.1
CPU	full_context	1573	1.00	7600.3	180.6	1249.1
GPU	pra_only	63	1.00	102.9	99.1	788.3
GPU	prefix_plus_pra	361	1.00	115.0	98.5	814.6
GPU	full_context	1573	1.00	1129.6	98.2	501.3

Controls recover the answer in 0/10 requests across CPU and GPU; every evidence-bearing condition succeeds in 30/30. Selected text uses 63 tokens, 96.0% fewer than the 1,573-token full context. On CPU, cold TTFT falls from 7,600.3 ms for full context to 549.2 ms for selected text. On GPU it falls from 1,129.6 to 102.9 ms. Prefix caching changes the comparison: warm full-context TTFT is 180.6 ms on CPU and 98.2 ms on GPU, versus 336.8 and 99.1 ms for selected text. Thus selected text strongly reduces cold ingestion, while a fully reusable prefix can remove most repeated-prefill cost in this fixed one-resource smoke.

Pipeline construction is reported separately: 1.58 s for the CPU one-shot process and 39.40 s for the first GPU one-shot process. The later direct continuous-batching GPU process constructs in 5.05 s after system/model caches have been populated; these are cold-process observations, not comparable steady-state request latencies.

5.3 Continuous batching

With one shared resource and concurrency eight, selected text reaches 8.48 requests/s compared with 7.39 for full context, while TTFT p99 falls from 297.7 to 148.9 ms. For independent resources, selected text still lowers TTFT p99 (262.0 versus 319.8 ms) but reaches only 5.88 requests/s versus 7.07. This small reversal prevents a blanket throughput claim. It may reflect output length, scheduler packing, and churn in the explicitly primed independent set; larger cohorts and matched generated-token work are required.

RSS spans roughly 1.95–2.19 GiB across endpoint rows and is not monotonic with transport or concurrency. The current experiment therefore supports a token and latency claim, not an engine-memory saving claim.

Table 3: Explicitly primed GPU continuous-batching endpoints. Every row has controlled answer success 1.00. RSS is process resident memory after the group.

Transport / resources	C	req/s	TTFT p99	generation p99	RSS MiB
selected / shared	1	1.35	159.7	791.6	2140
selected / shared	8	8.48	148.9	957.8	1950
selected / independent	1	1.57	115.6	760.3	1960
selected / independent	8	5.88	262.0	1430.8	2147
full / shared	1	1.55	223.7	832.1	2163
full / shared	8	7.39	297.7	1154.7	2115
full / independent	1	1.46	188.3	850.3	2119
full / independent	8	7.07	319.8	1192.3	2193

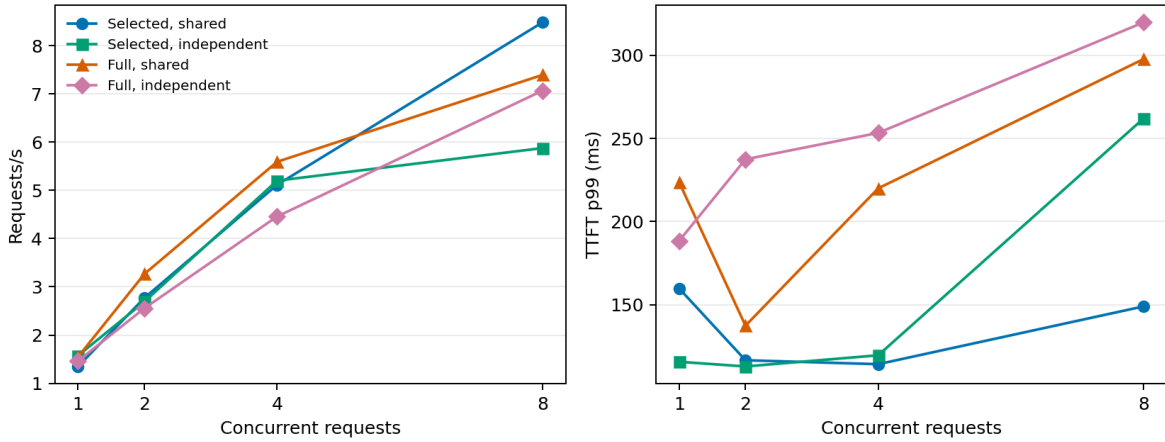


Figure 1: Warm OpenVINO GenAI E0 throughput and TTFT tail under direct continuous batching. Selected denotes rendered selected text, not native E2 K/V.

6 Interpretation

OpenVINO prefix caching and PRA selection address different reuse axes. When the same full context is repeatedly visible, prefix caching can make its warm TTFT competitive with selected text. When requests arrive concurrently, shorter selected inputs reduce scheduler work and improve the shared-resource frontier in this smoke. Native PRA would target a harder regime: encode a resource once, select different non-prefix intervals for many queries, and avoid making those source tokens visible. That regime cannot be measured until the attention attachment seam exists.

7 Required Native Extension

A minimal extension should accept an authorized set of topology-matched K/V handles plus source positions at request admission. The scheduler must pin the set, keep it outside ordinary prefix identity, expose it once to attention, and release it on completion or cancellation. Attention must normalize local and memory scores jointly. A geometry ladder should then compare ordinary prefix, geometry-matched cached source, full native source, sparse multi-record source, and cached decode before any economics claim.

8 Limitations

The controlled task verifies evidence transport, not natural QA. Only one small INT4 model, one CPU, one integrated GPU, and five waves are measured. There is no OVMS process, NPU, native K/V parity, device-memory telemetry, long-session pressure, eviction sweep, or natural QASPER/Hotpot/2Wiki cohort. All serving results are E0.

9 Conclusion

PRA can coexist with OpenVINO’s public serving abstractions at E0 while preserving semantic identity and explicit capability reporting. Selected text substantially reduces cold ingestion and improves the shared-resource continuous-batching frontier on the tested GPU. OpenVINO GenAI exposes the physical scheduling controls needed around a future integration but not the public arbitrary non-prefix K/V attachment needed for E2. That boundary is now implemented as a safe contract and measured as an open gate.

Reproducibility

Raw JSON, generated tables and plots, environment audit, and runners are under `docs/papers/shared/results/paper6_3_openvino` and `experiments/paper6_3_openvino`.

References

- [1] Intel. *OpenVINO GenAI LLMPipeline and SchedulerConfig API Reference*. https://docs.openvino.ai/2026/api/genai_api/
- [2] OpenVINO Toolkit. *OpenVINO GenAI*. <https://github.com/openvinotoolkit/openvino.genai>